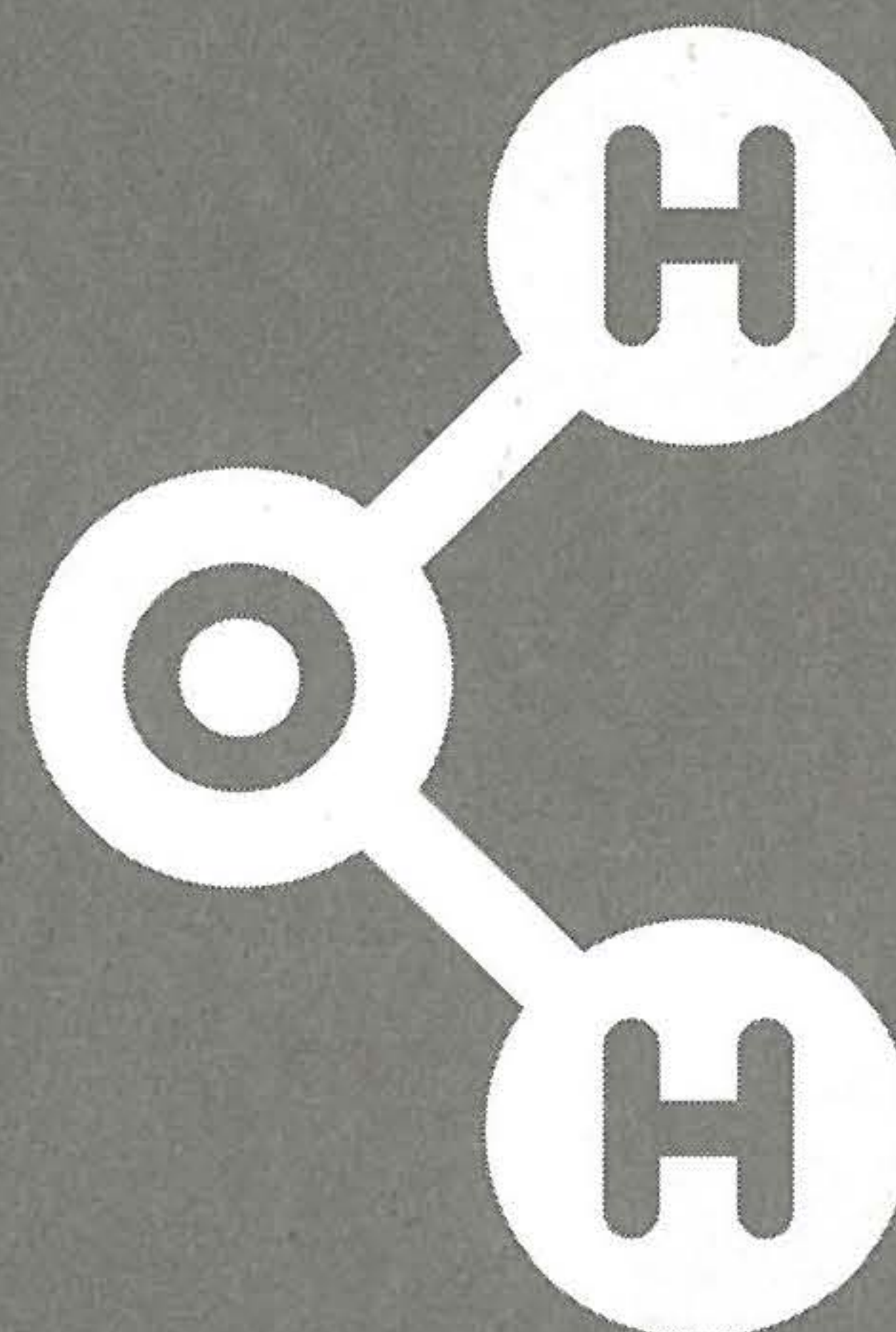




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PAPER A



2011
ICAS

International Competitions
and Assessments for Schools

SCIENCE

Educational
Assessment
Australia

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

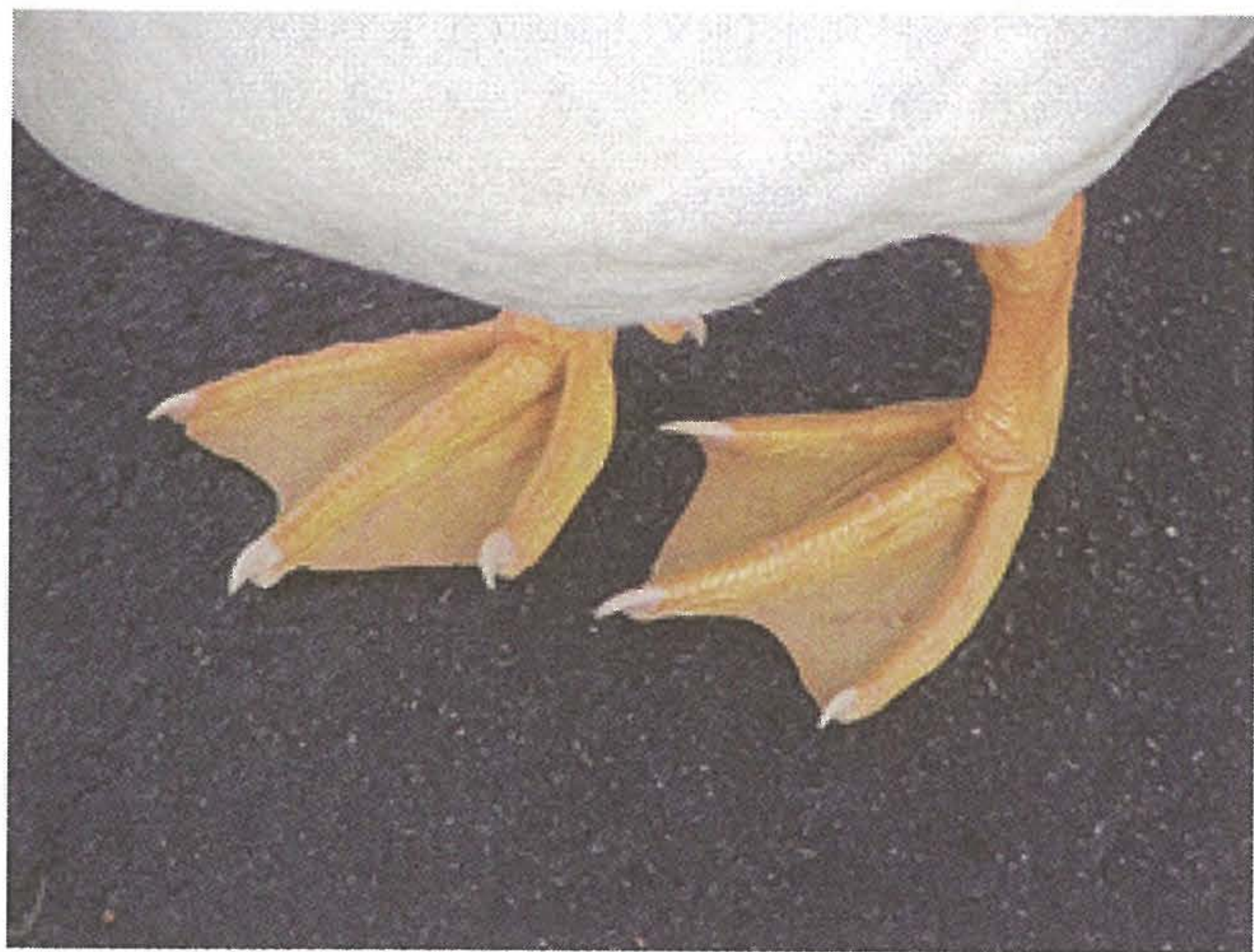
There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

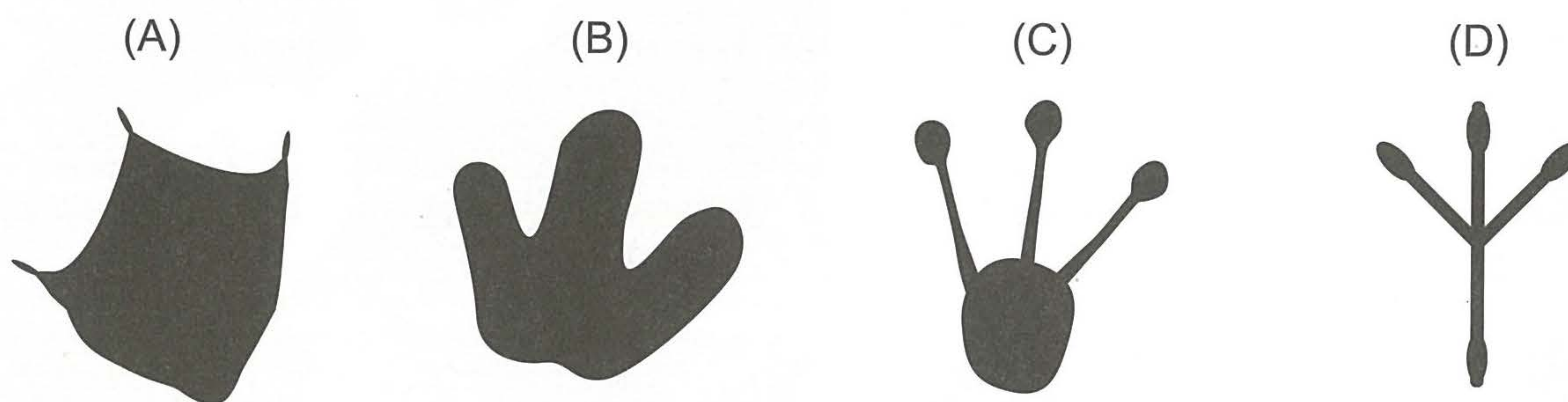
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

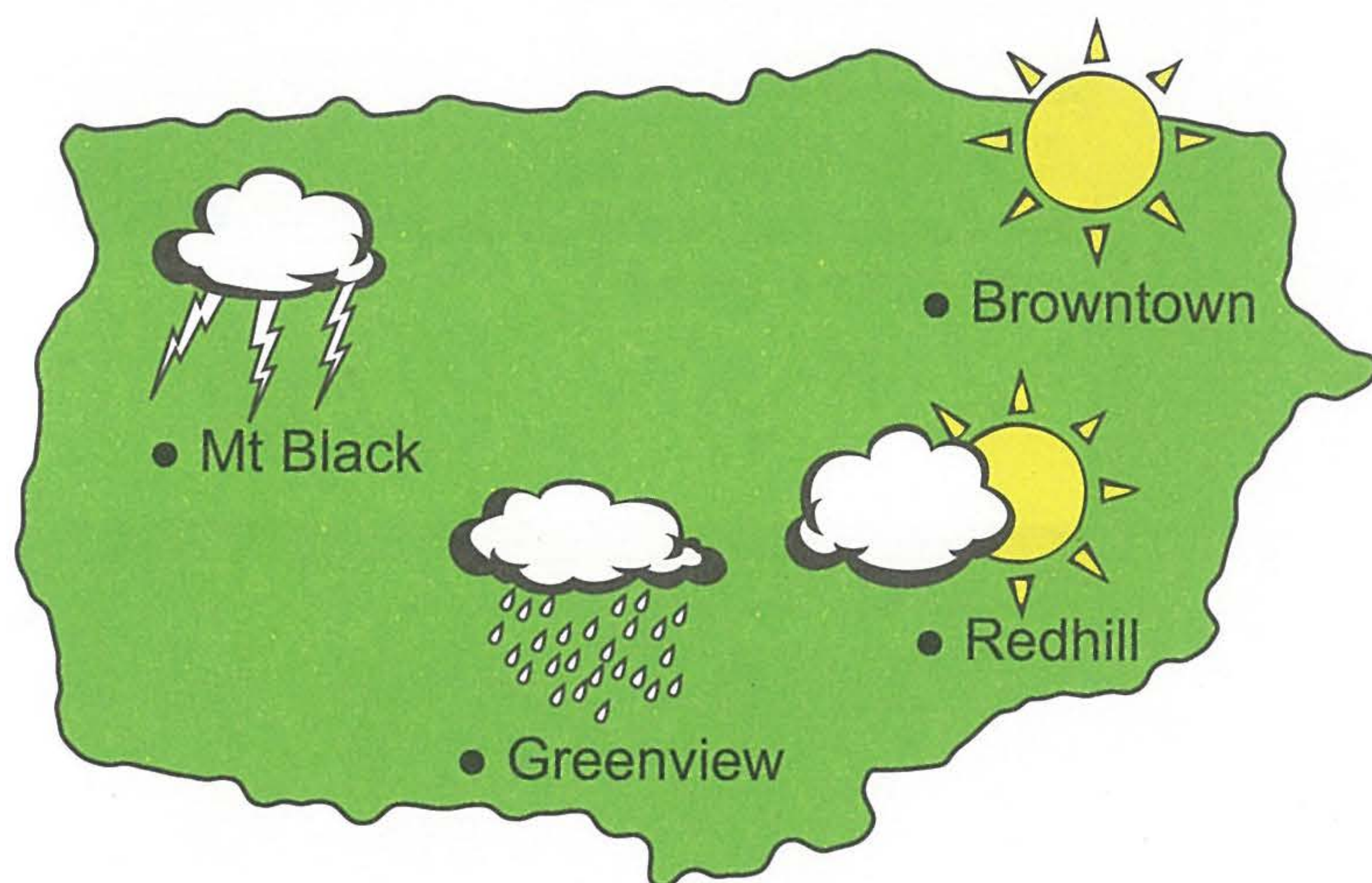
1. This is a picture of a duck's feet.



Which footprint was left by this duck?



2. The diagram shows the weather in some towns.

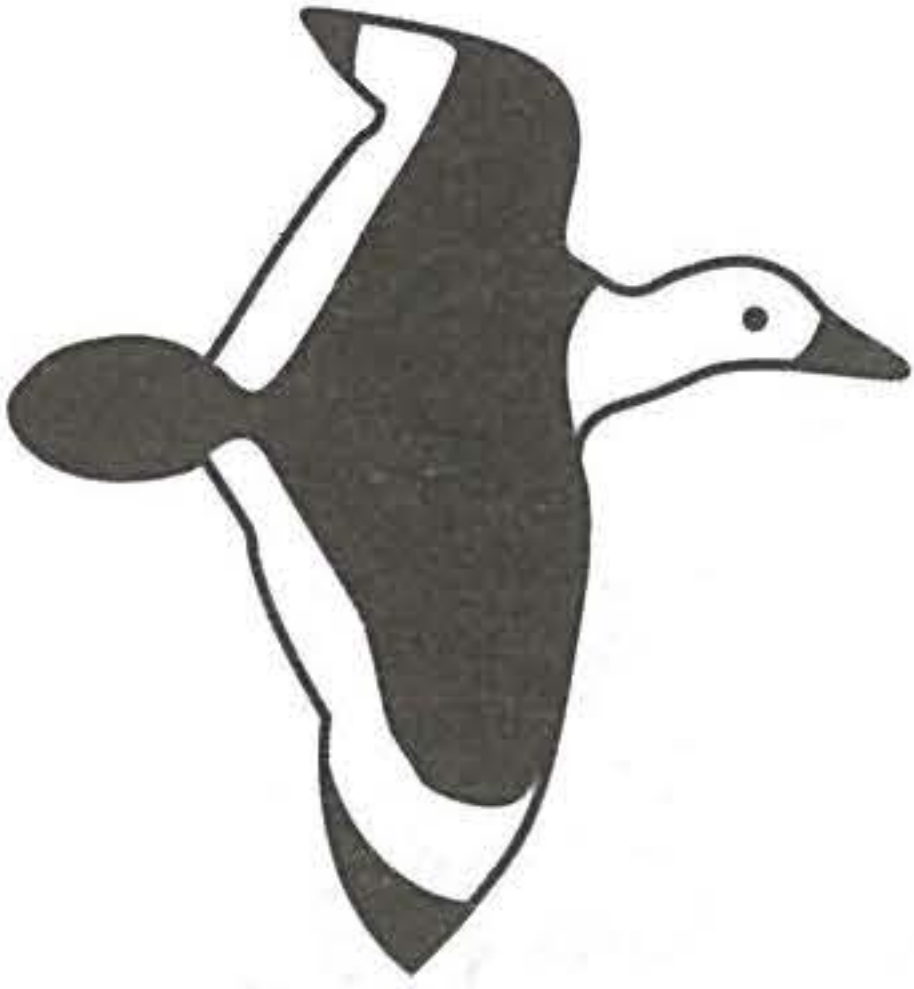
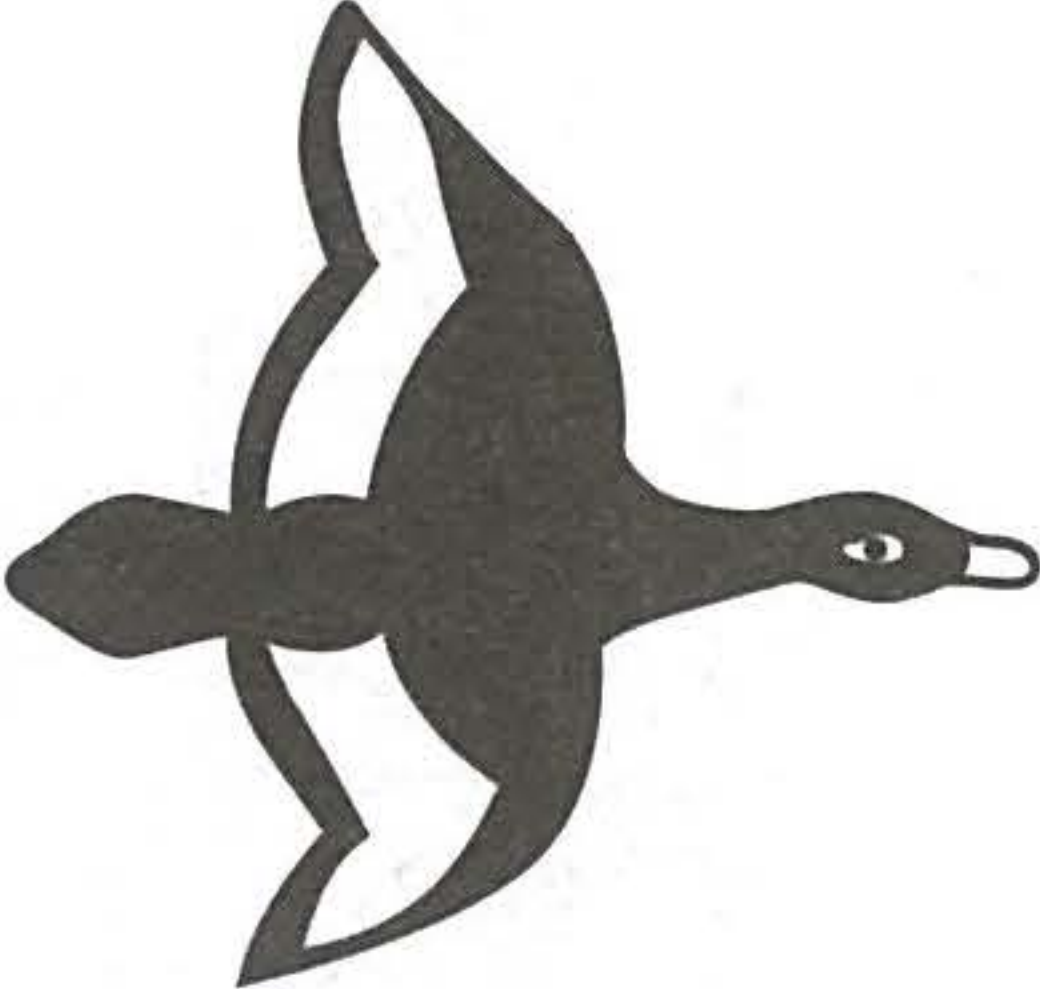


KEY	
	thunderstorms
	rainy
	partly cloudy
	sunny

Which town has sunny weather?

- (A) Mt Black
(B) Browntown
(C) Redhill
(D) Greenview

3. Kelsey drew three different flying birds.

Bird 1	Bird 2	Bird 3
		

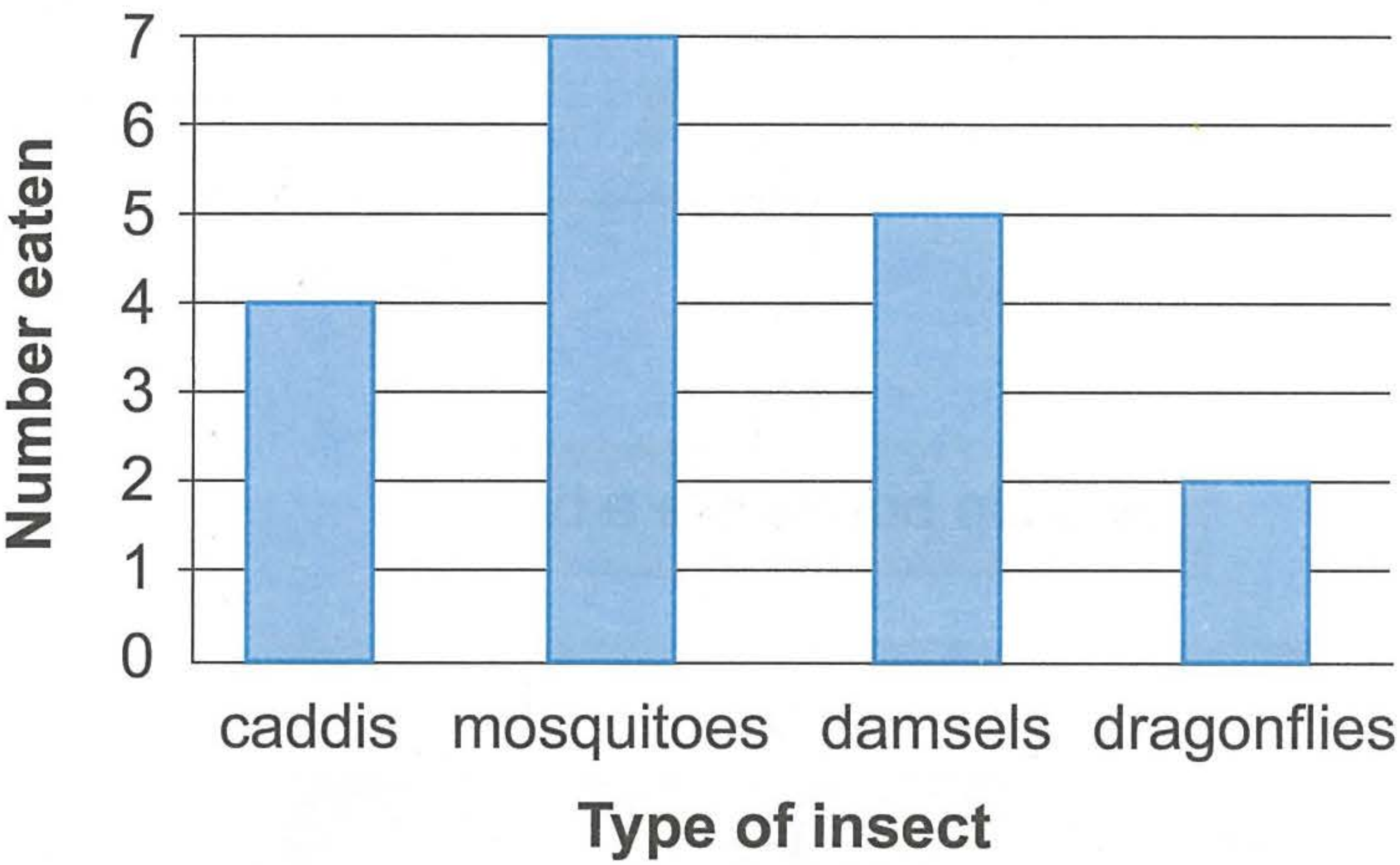
Which feature belongs only to Bird 1?

- (A) white head
- (B) pointy beak
- (C) white stripe on its wings
- (D) upright feathers on its head

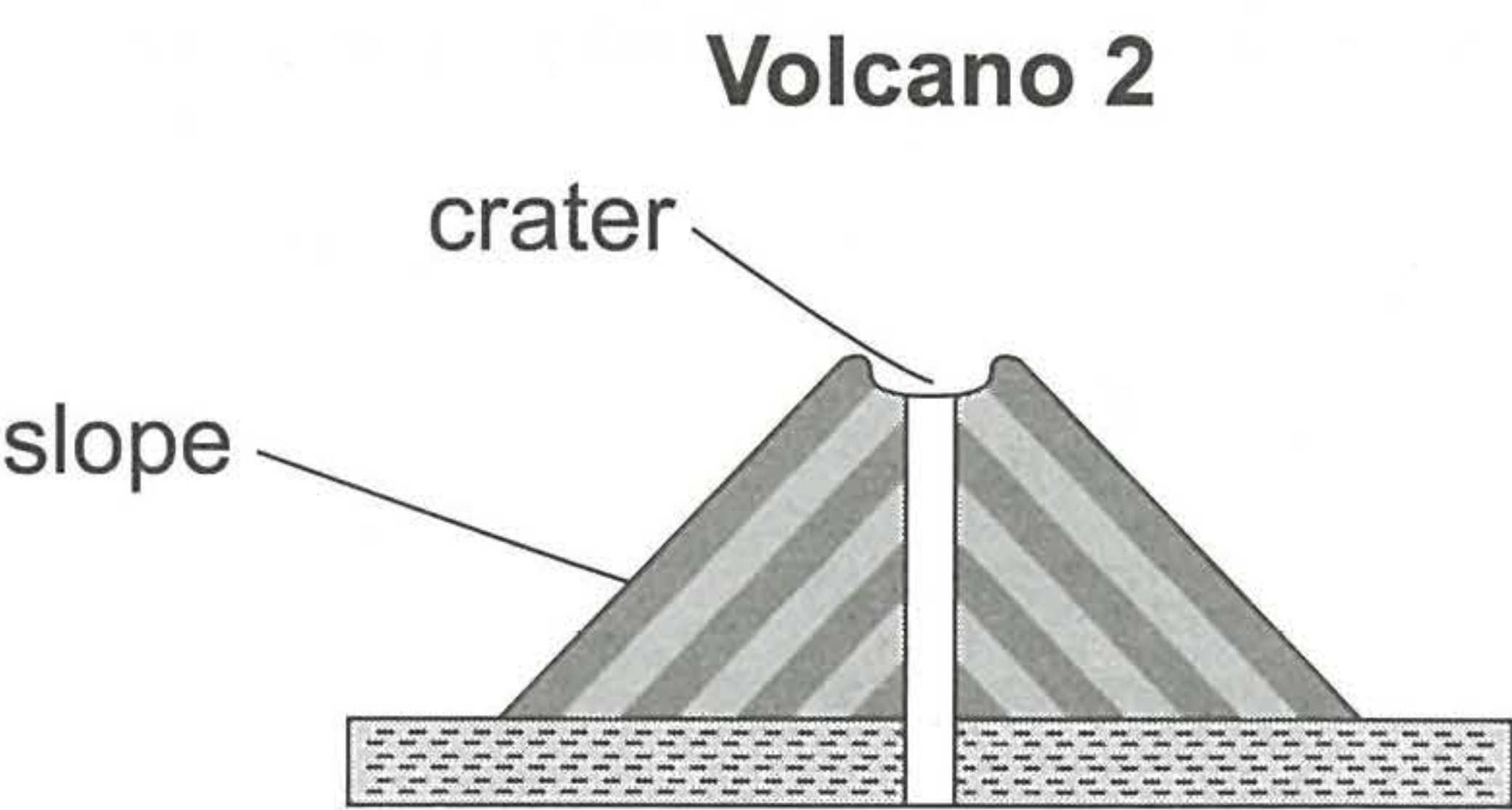
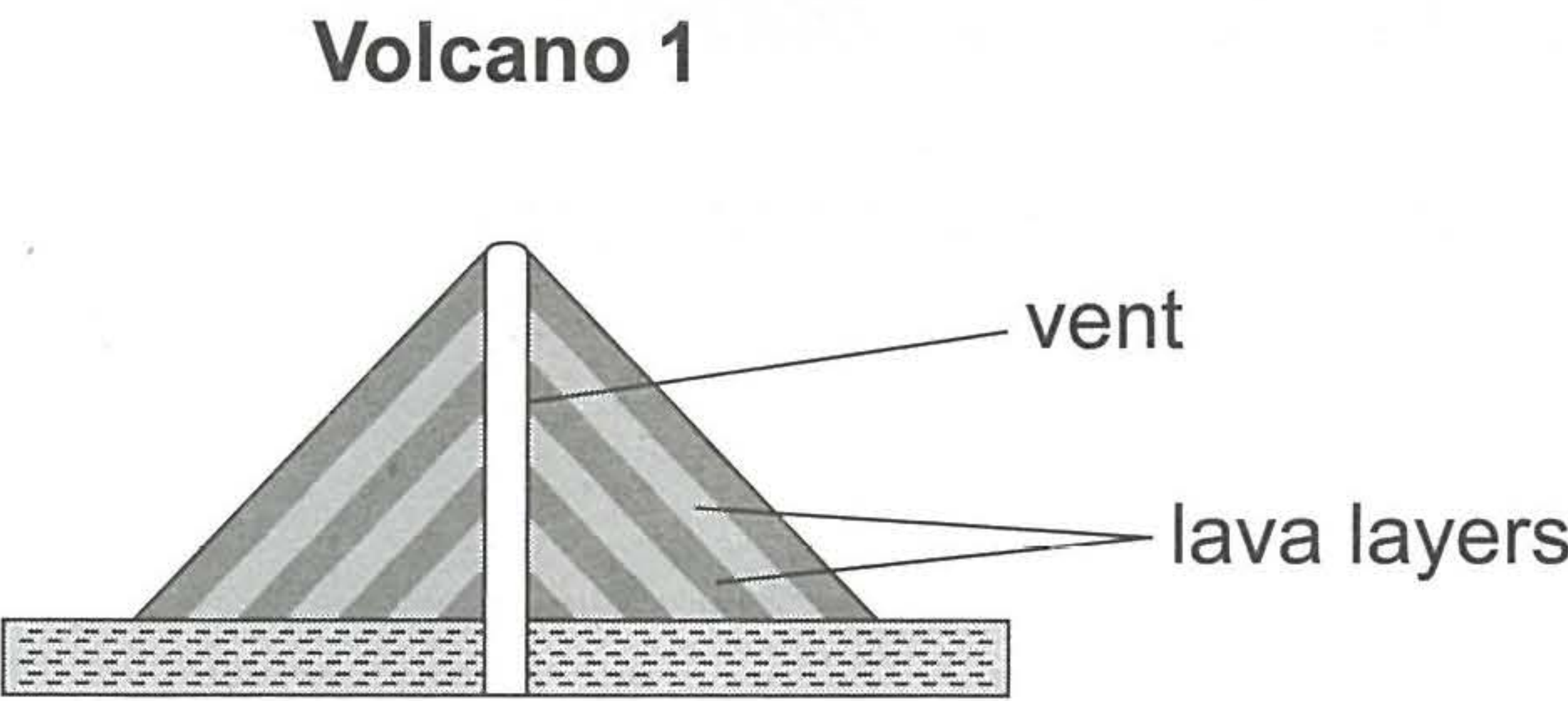
4. The graph shows the number of different insects eaten by a fish.

How many damself did the fish eat?

- (A) 2
- (B) 4
- (C) 5
- (D) 7



5. Marty drew these diagrams of two volcanoes.

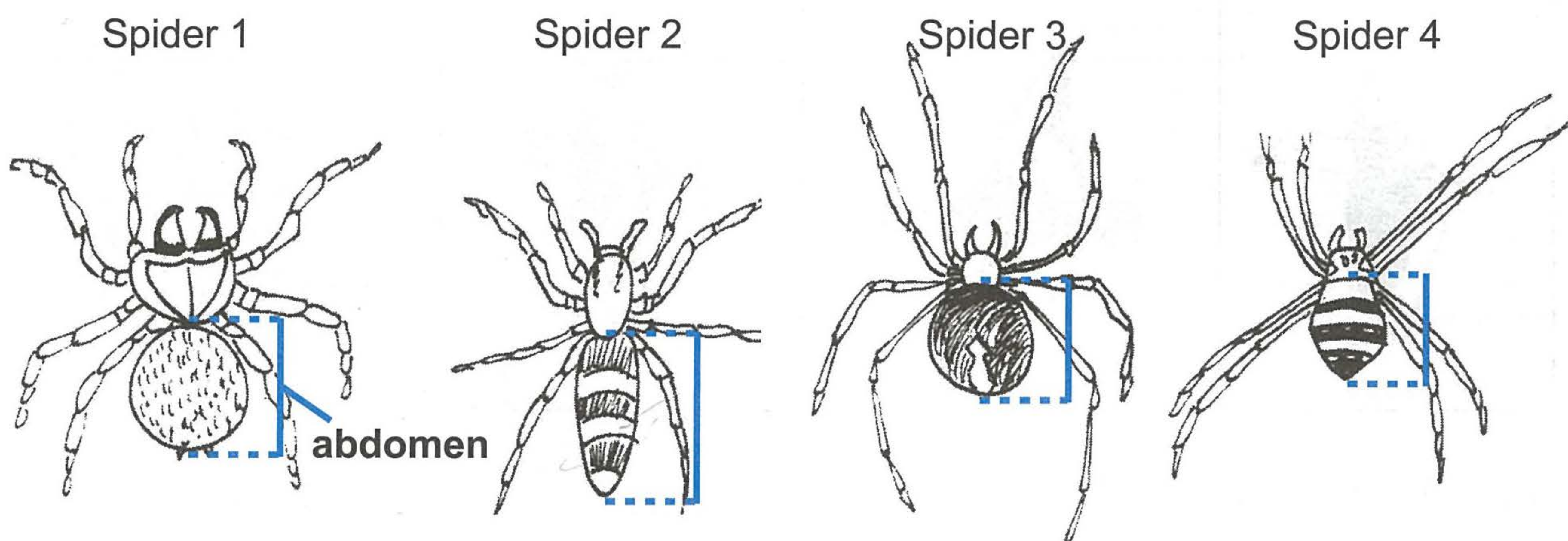


What is different about these volcanoes?

- (A) the crater at the top
- (B) the location of the vent
- (C) the slope of the sides
- (D) the number of lava layers

6. Peta drew four spiders. She labelled the abdomen on each spider.

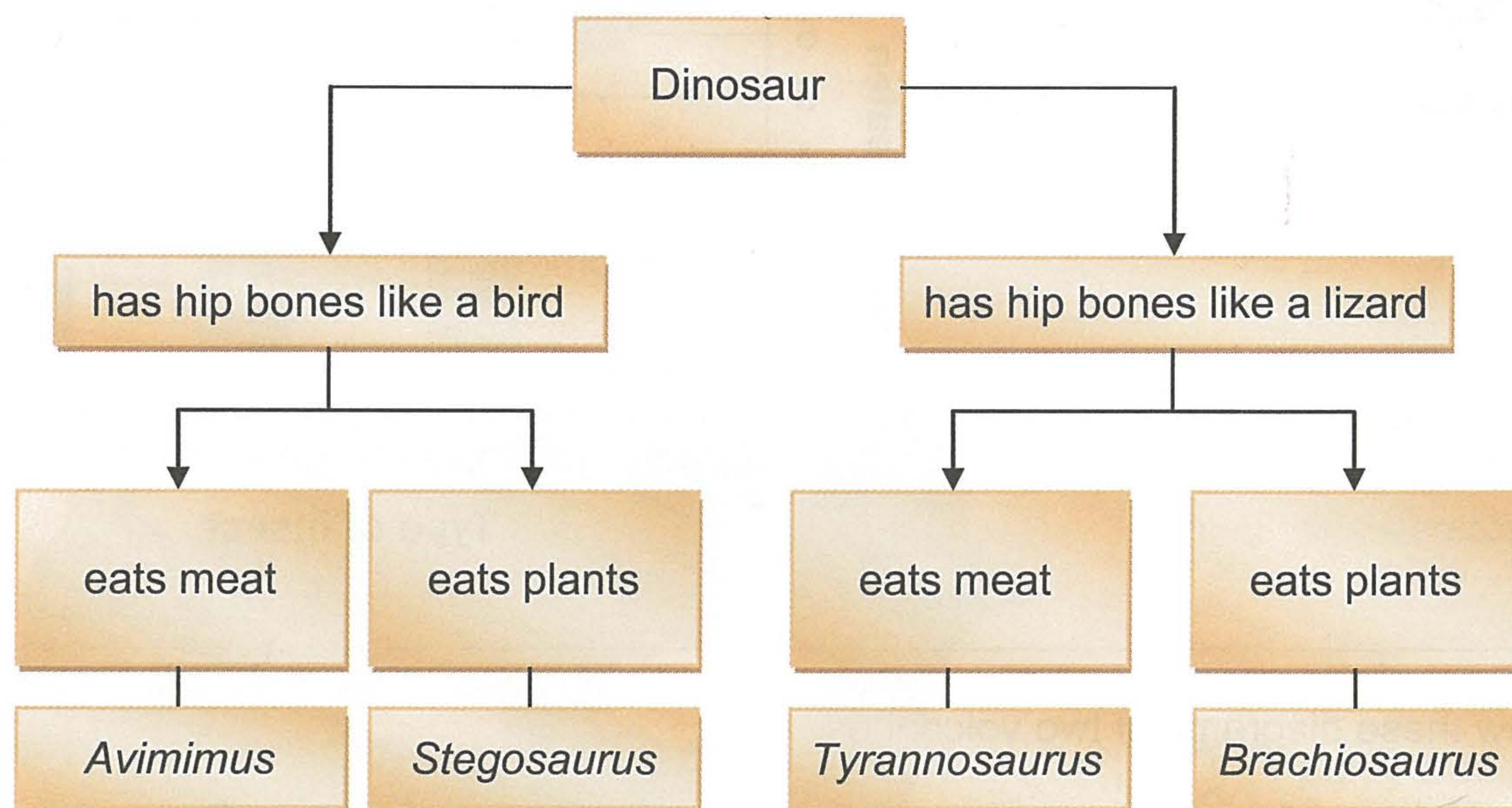
The drawings show the spiders at the same scale.



Which spider has the longest abdomen?

- (A) Spider 1 (B) Spider 2 (C) Spider 3 (D) Spider 4

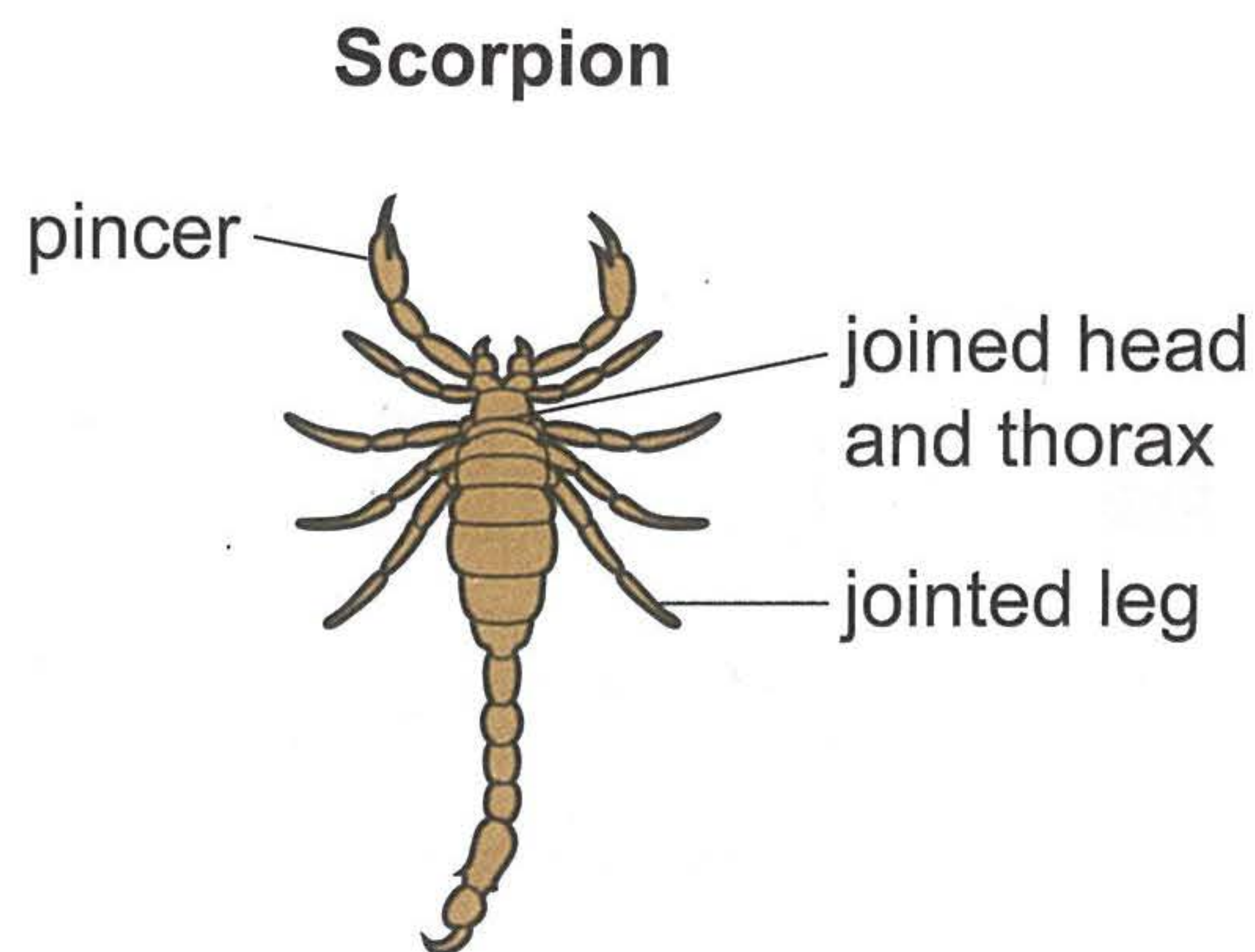
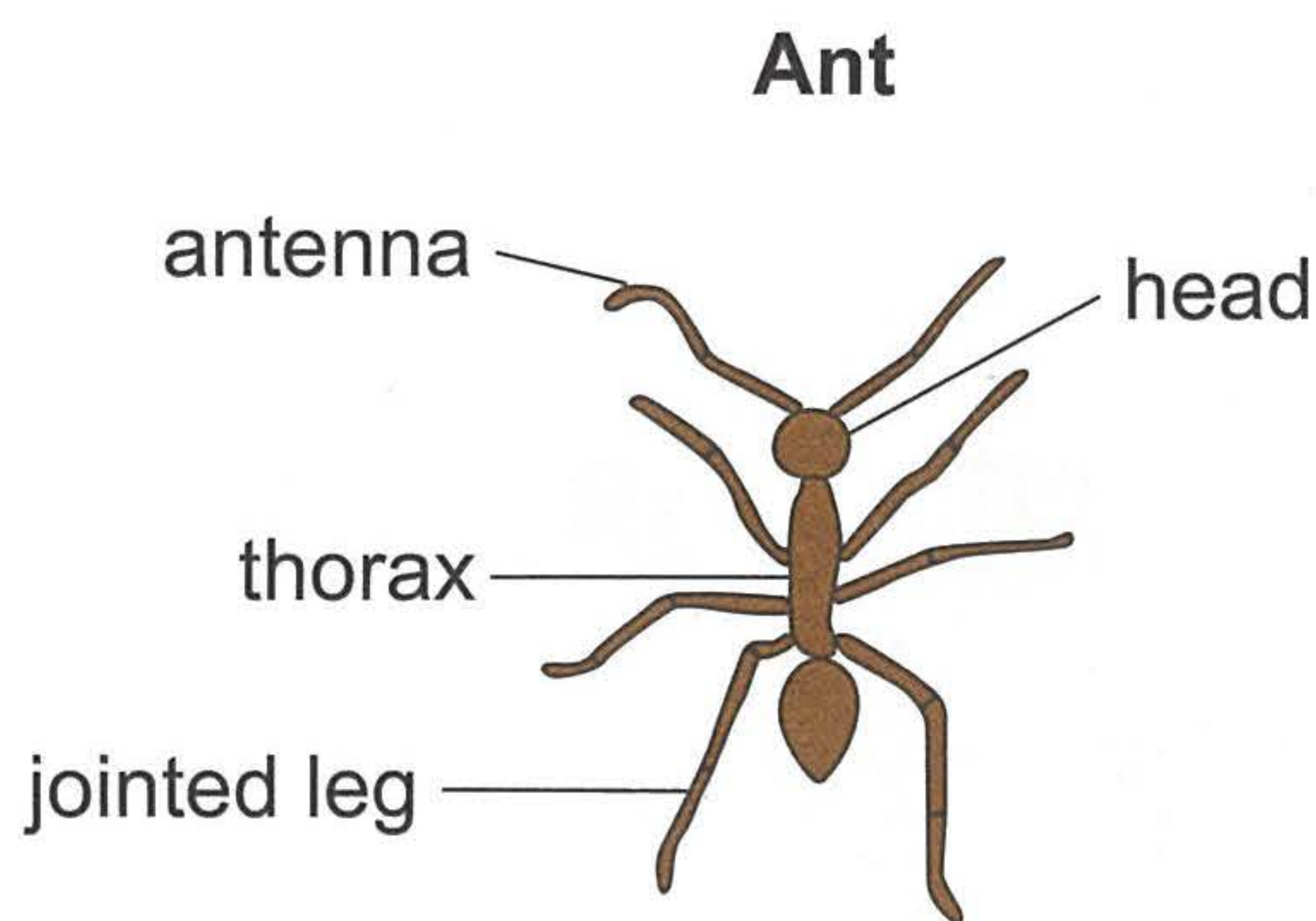
7. James drew this diagram about dinosaurs.



Using this diagram, which dinosaur has hip bones like a lizard and eats meat?

- (A) *Avimimus*
(B) *Stegosaurus*
(C) *Tyrannosaurus*
(D) *Brachiosaurus*

8. Here are pictures of two small animals.



Both of these animals have

- (A)

six legs.
- (B)

pincers.
- (C)

jointed legs.
- (D)

two antennae.

9. The diagram shows how water can enter and leave a flowering plant.

What part of the plant carries water from the roots to the leaves?

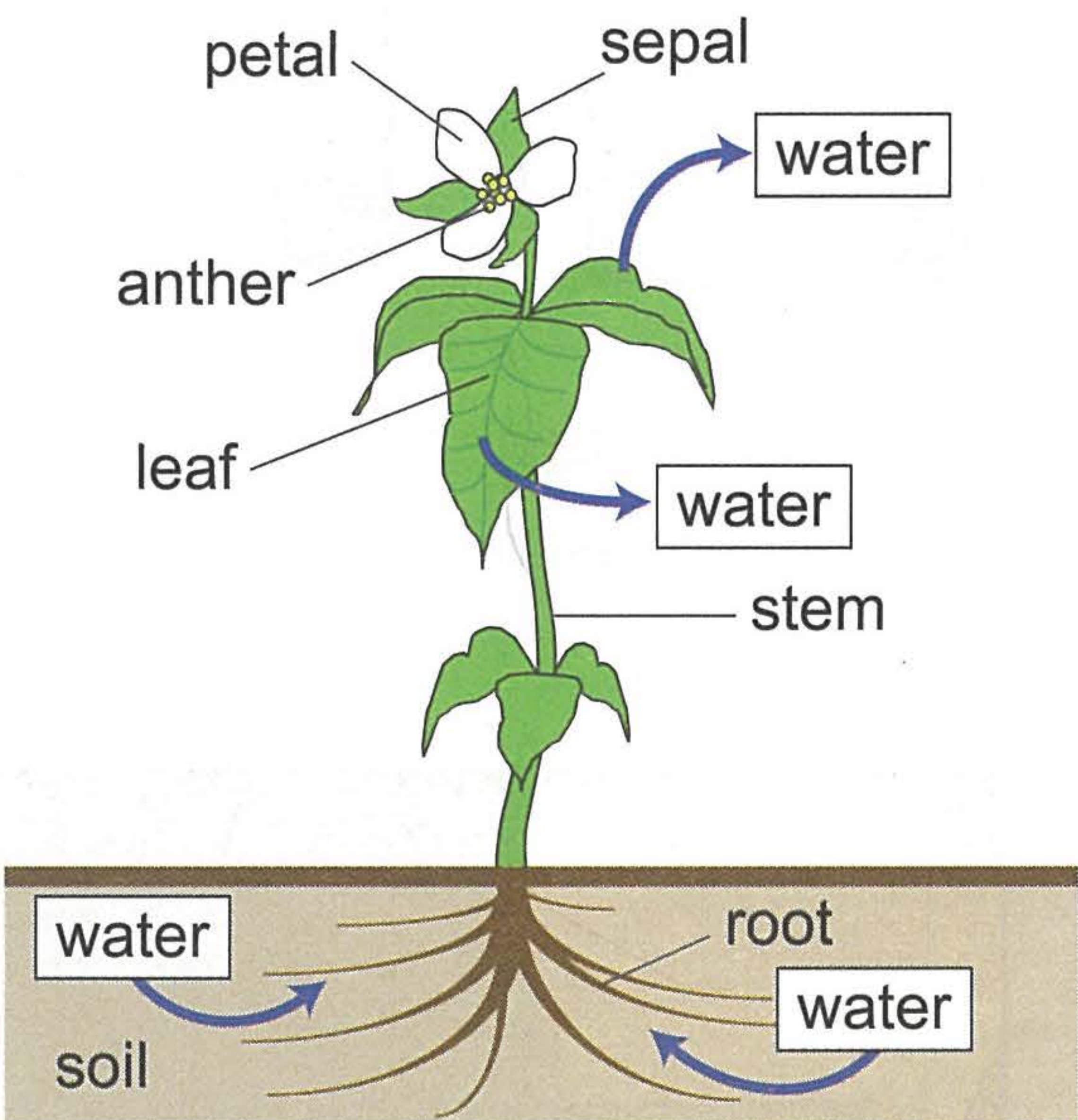
- (A)

stem
- (B)

petal
- (C)

sepal
- (D)

anther



10. The table shows weather events that occurred at four different places in 2010.

Place	Weather events in 2010				
	Snow storm	Thunderstorm	Cyclone	Flood	Heatwave
Portville	✓	✓			✓
New City					✓
Rivertown		✓	✓	✓	
Bulla Village		✓	✓		

Which two places had the most weather events in common?

- (A)

Portville and New City
- (B)

Portville and Rivertown
- (C)

New City and Bulla Village
- (D)

Rivertown and Bulla Village

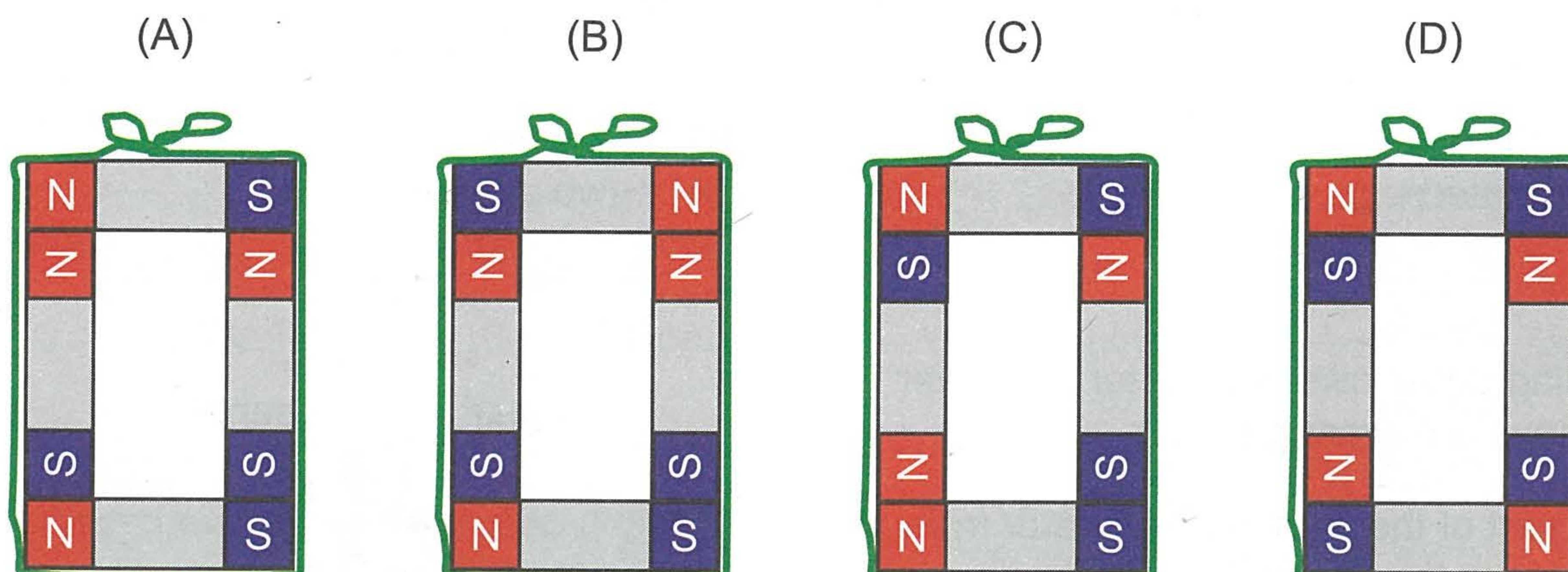
11. Magnets have a north end (N) and a south end (S). When two magnets are brought together end to end, they push or pull each other depending on which ends are together.

The diagram shows what happens.

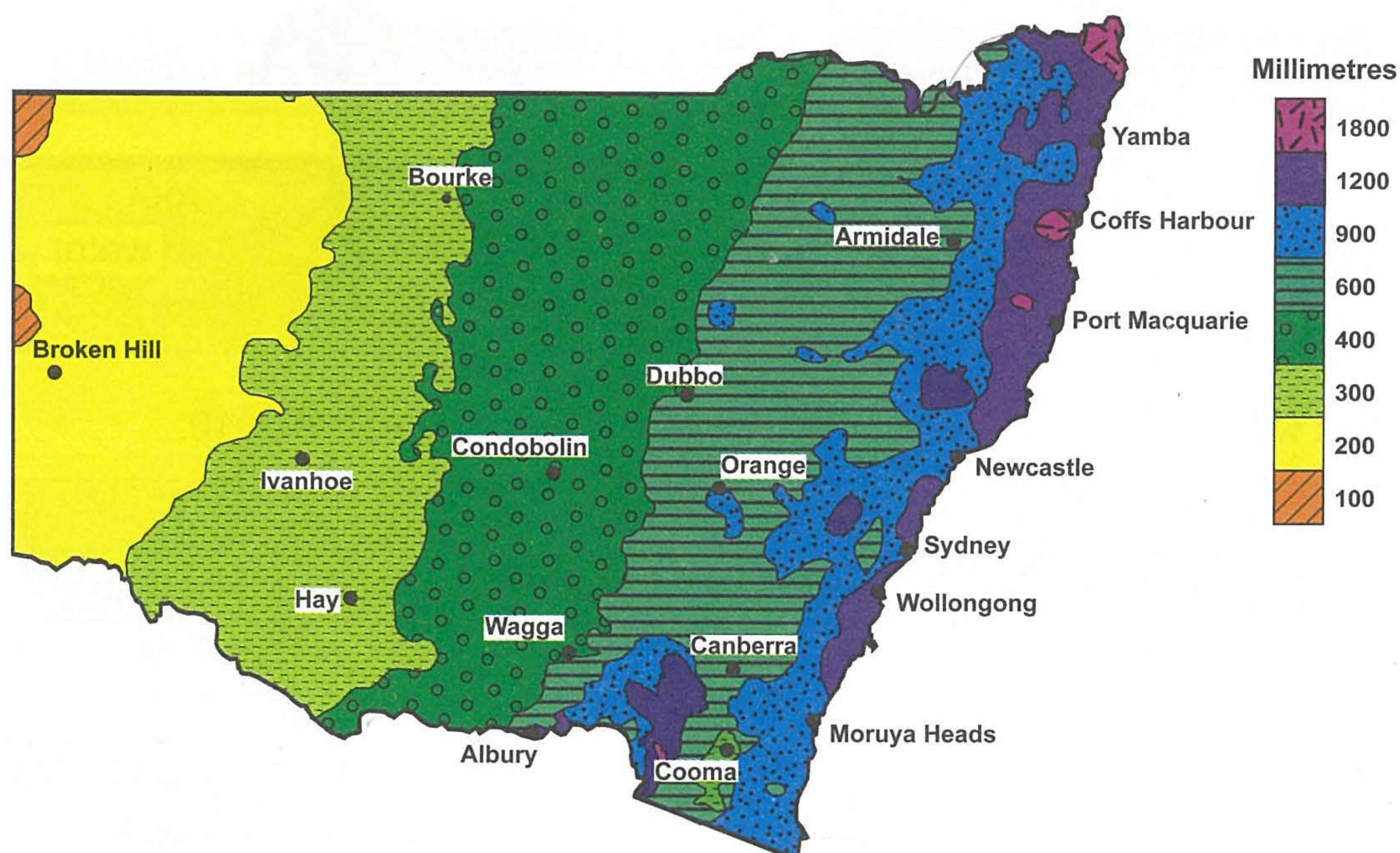


Here are some groups of magnets tied together by string.

Which group will stay together when the string is taken away?



12. This map shows the annual rainfall across New South Wales.



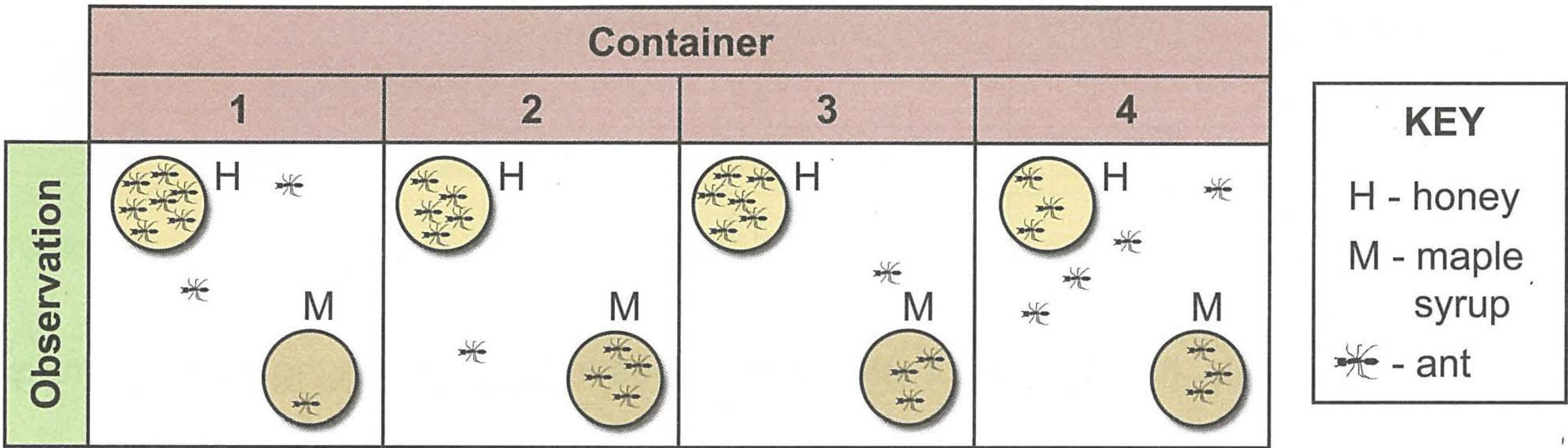
Which of these towns receives approximately 900 millimetres of rainfall annually?

- (A) Dubbo (B) Orange
(C) Wollongong (D) Coffs Harbour

13. Jen knew that ants like sweet food. She wondered if they prefer honey or maple syrup.

In four identical containers, she placed a dish with 5 g of honey and a dish with 5 g of maple syrup. She then placed 10 ants in the middle of each container.

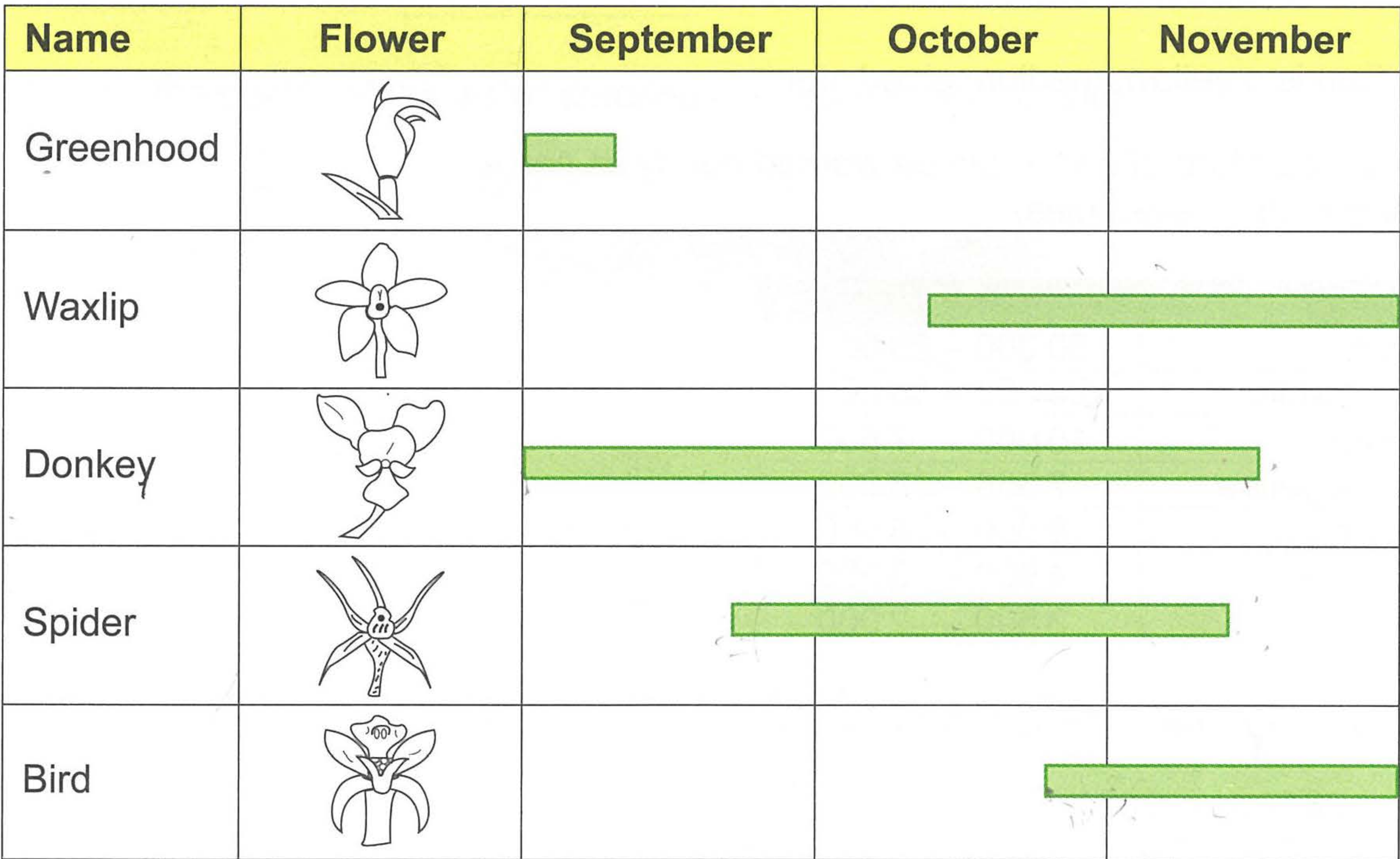
Here are her observations after 20 minutes.



From these observations, which statement is true?

- (A) All of the ants preferred honey.
- (B) The ants did not like maple syrup.
- (C) More ants preferred honey to maple syrup.
- (D) The ants liked honey and maple syrup equally.

14. Orchids are flowering plants. Kim drew this chart to show when five different types of orchids flowered in the bushland near her house.



Which three types of orchids flowered together for the longest time?

- (A) Waxlip, Donkey and Spider

(B) Waxlip, Donkey and Bird
- (C) Greenhood, Donkey and Waxlip

(D) Greenhood, Donkey and Bird

For questions 15 and 16 use the information below.

Some students measured how long it took carrot seeds to sprout at different temperatures.

The table shows their results.

Temperature (°C)	Days taken to sprout
22	17
24	15
26	13

15. Based on the table, how long will it take carrot seeds to sprout when the temperature is 28°C?

- (A) 9 days (B) 11 days
(C) between 13 and 17 days (D) more than 17 days

16. Which idea could the students have been testing in this experiment?

- (A) Carrot seeds grow faster in pots.
(B) Carrot seeds grow at any temperature.
(C) Carrot seeds take the same time to sprout at different locations.
(D) Carrot seeds take the same time to sprout at different temperatures.

17. The Sun is a yellow, medium-sized star.

The temperature of a star can be worked out by its colour.
Here are some examples:

Colour	Temperature (°C)
blue	50 000 – 28 000
blue-white	28 000 – 10 000
white	10 000 – 7 500
white-yellow	7 500 – 6 000
yellow	6 000 – 4 900
orange	4 900 – 3 500
red	3 500 – 2 000

Based on this information, what is the temperature of the Sun?

- (A) 28 000 – 10 000 °C
(B) 10 000 – 7 500 °C
(C) 7 500 – 6 000 °C
(D) 6 000 – 4 900 °C

18. The table lists information about some food substances found in a kitchen.

Substance	Appearance	Ingredients
P	powder	salt, flour, sugar, yeast extract
Q	crystals	sugar, gelatine, flavour, colour
R	paste	yeast extract, salt, malt extract, colour
S	powder	sugar, cocoa, milk solids, malt extract
T	paste	peanuts, canola oil, sugar, salt

What makes substance **R** different from all the other substances?

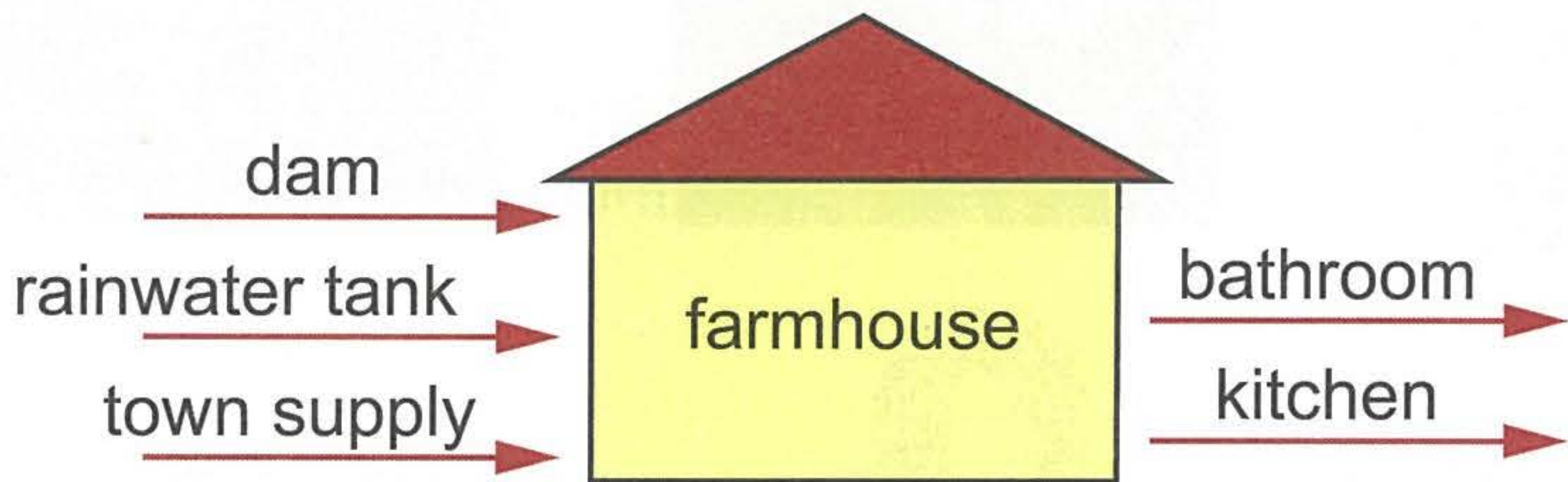
- (A) It is a paste.

(B) It contains salt.
- (C) It does **not** contain sugar.

(D) It does **not** contain yeast extract.

For questions 19 and 20 use the information below.

The diagram shows the water that enters and leaves Sam’s farmhouse. The arrows show the direction of the water flow.



Sam tested the quality of the water entering and leaving her farmhouse.

The table shows Sam’s test results.

Water from	Colour	Taste /smell	Water clarity
dam	brown	earthy	clear
rainwater tank	colourless	none	clear
town supply	colourless	earthy	clear
bathroom	grey	soapy	cloudy
kitchen	grey	soapy	cloudy

19. What colour is the water that flows out of the farmhouse?

- (A) grey

(B) brown
- (C) clear

(D) cloudy

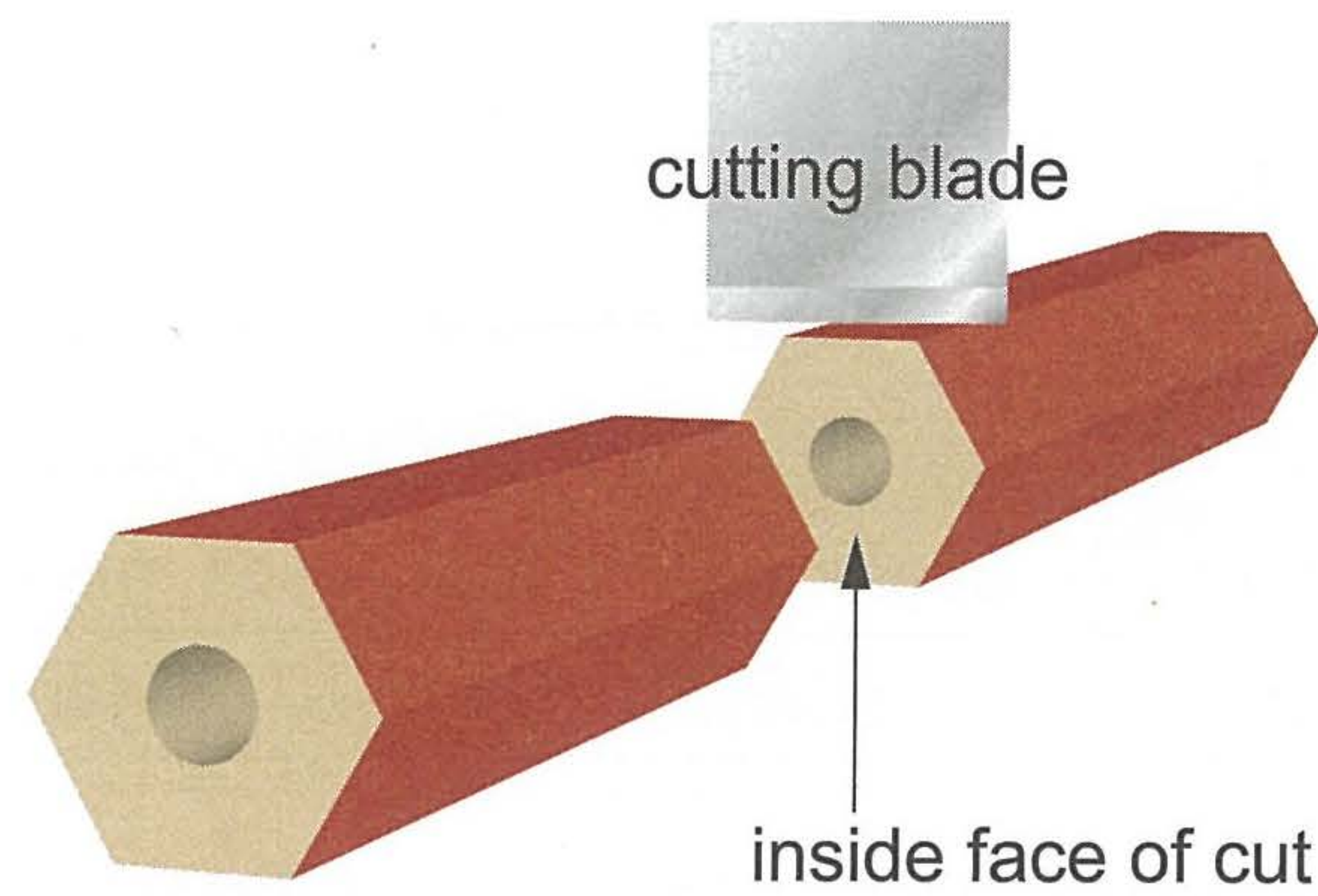
20. Using the test results, where did the best quality water for drinking come from?

- (A) dam

(B) kitchen
- (C) town supply

(D) rainwater tank

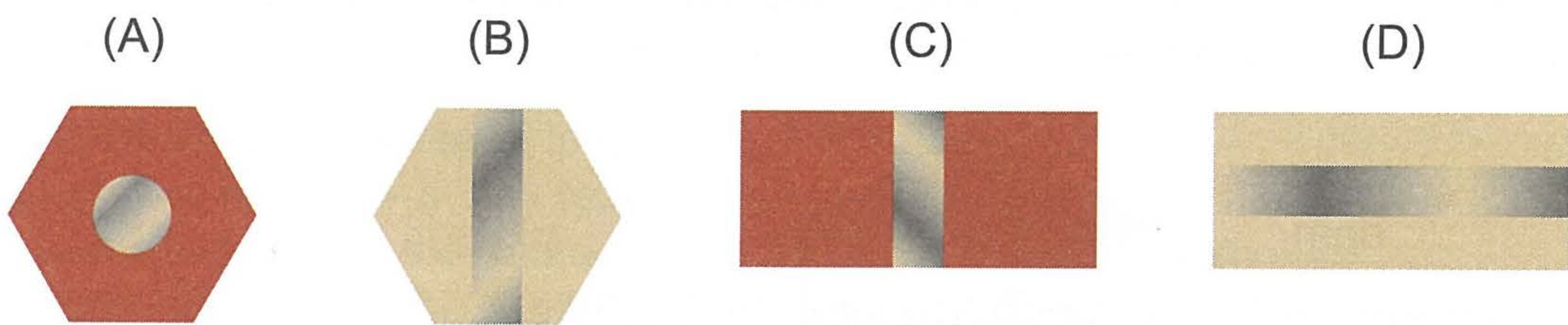
21. A pencil has been cut in half. This diagram shows what it looks like inside the cut.



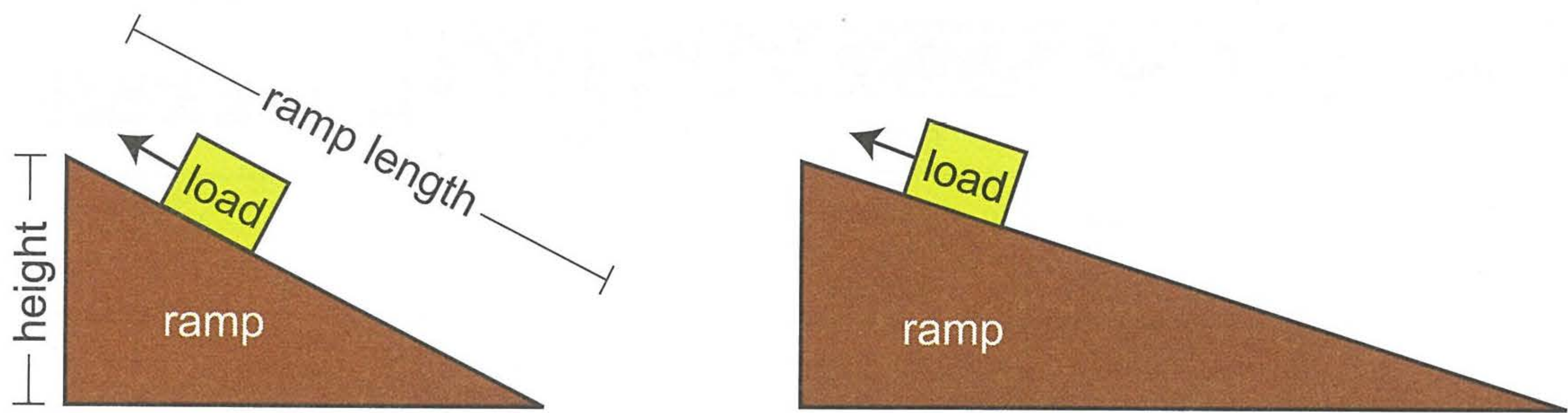
Another pencil was cut in half as shown in the diagram.



What will the inside face of the cut look like?



22. Tim made two ramps from the same type of wood. He used them to pull a load up a slope. The diagrams show his ramps.



What did Tim change in his investigation?

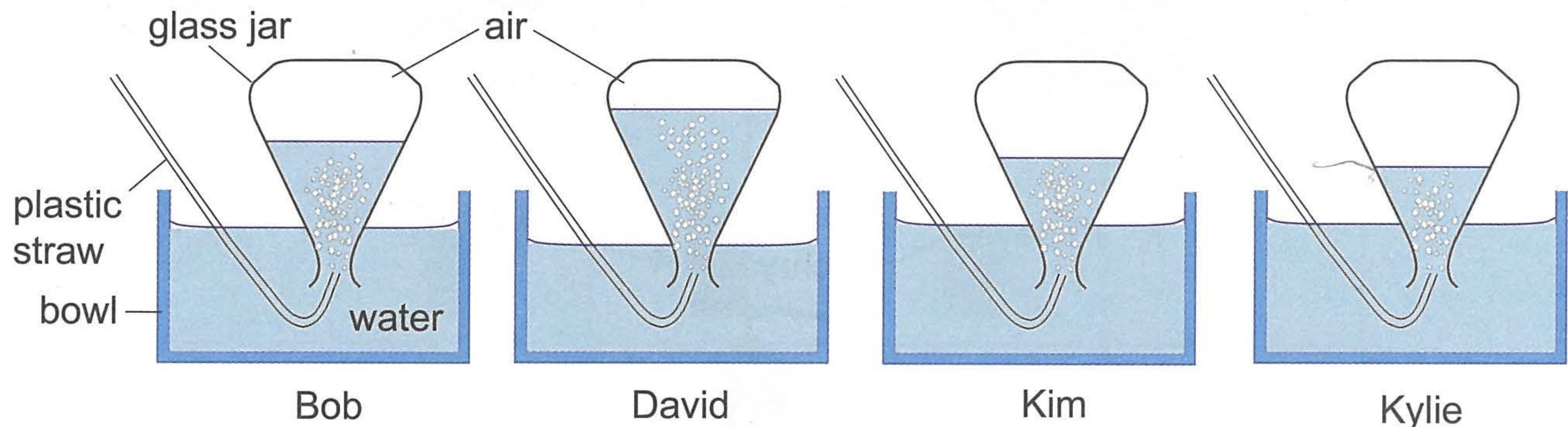
- (A) the smoothness of the ramp surface
- (B) the height the load was raised
- (C) the steepness of the ramp
- (D) the size of the load

23. Four children investigated the amount of air they could hold in their lungs.

Each child

- filled a glass jar completely with water.
- turned the jar upside down in a bowl of water without losing water from the jar.
- took the deepest breath possible and blew out nearly all of the air from their lungs into a curved straw.

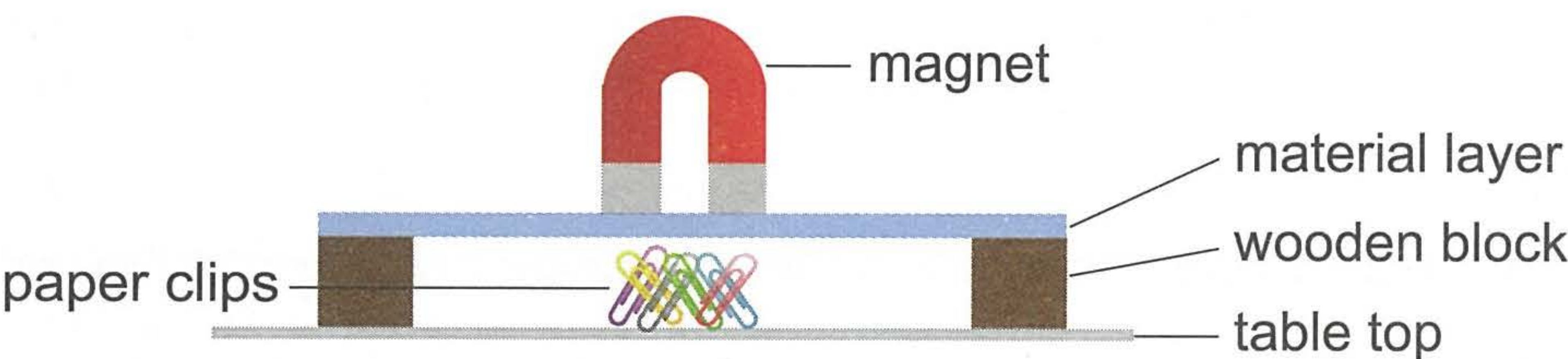
The drawing shows what they observed.



Which child held the most air in his or her lungs?

- (A) Bob (B) David (C) Kim (D) Kylie

24. Ben is planning an experiment using a magnet and steel paper clips. The diagram shows how he set up his equipment.



Here is the table he created to record his results before doing the experiment.

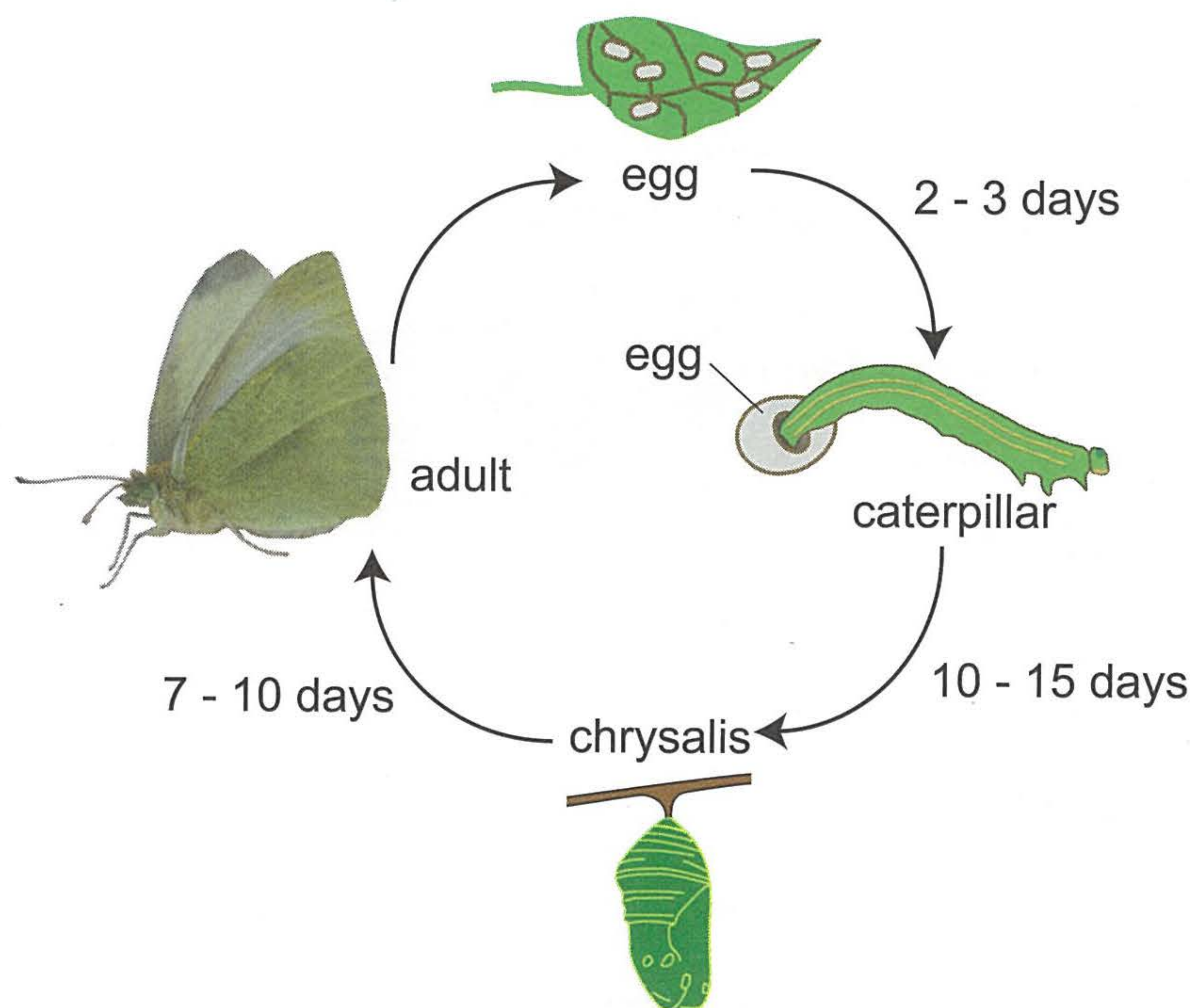
Test	Material	Thickness of material layer (mm)	Number of paper clips picked up
1	cardboard	2	
2	glass	4	
3	wood	6	
4	cardboard	4	
5	glass	6	
6	wood	8	

Ben wants to find out what happens when the thickness of the material layer is doubled.

Which two tests should Ben use?

- (A) Tests 1 and 2 (B) Tests 1 and 4
(C) Tests 2 and 5 (D) Tests 4 and 6

25. The diagram shows the life cycle of a butterfly from egg to adult.



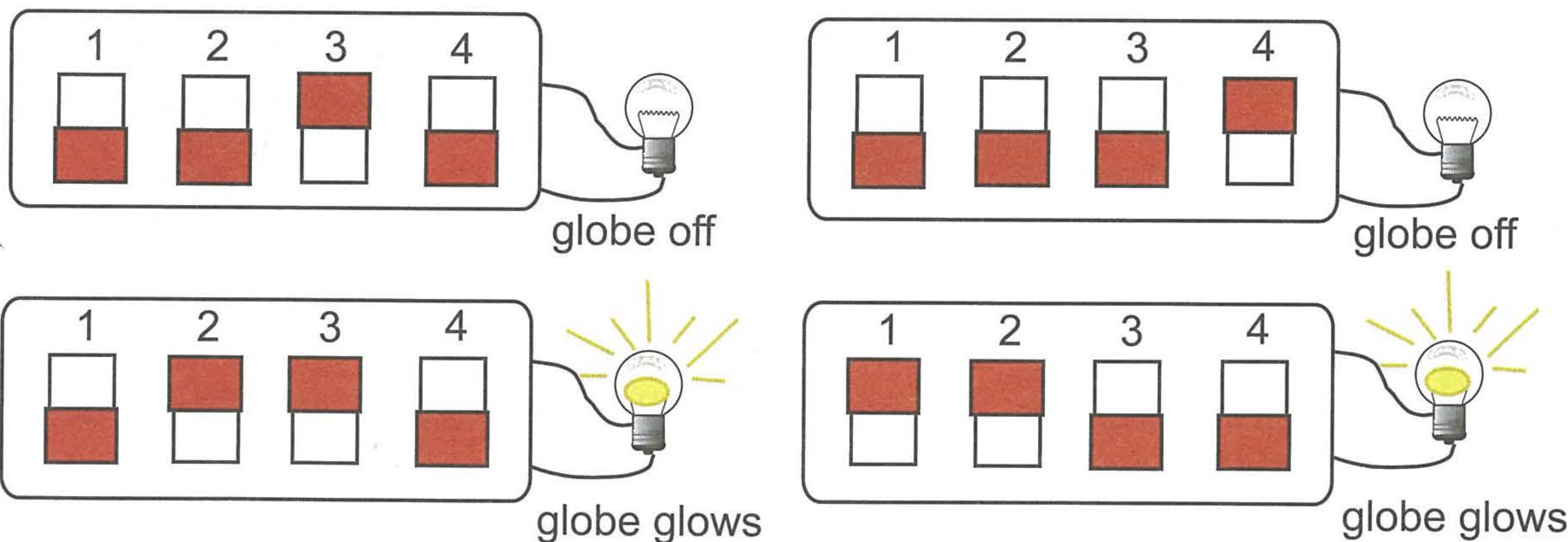
A butterfly can lay eggs as soon as it is an adult.

What is the smallest number of days after hatching that this butterfly would be able to lay eggs?

- (A) 2 (B) 7 (C) 17 (D) 19

26. When a switch closes an electrical circuit, the light globe belonging to the circuit will glow.

Emily did not know which switch closed the circuit belonging to a light globe. She used the tests shown to find out.



Which switch closed the circuit belonging to the light globe?

- (A) 1 (B) 2 (C) 3 (D) 4

27. Vitamin C changes a black starch solution to grey.

Mark wanted to find out which brand of fruit juice had the most vitamin C. He tested four brands.

Mark put equal amounts of black starch solution into four test tubes. He then added some fruit juice to each test tube.

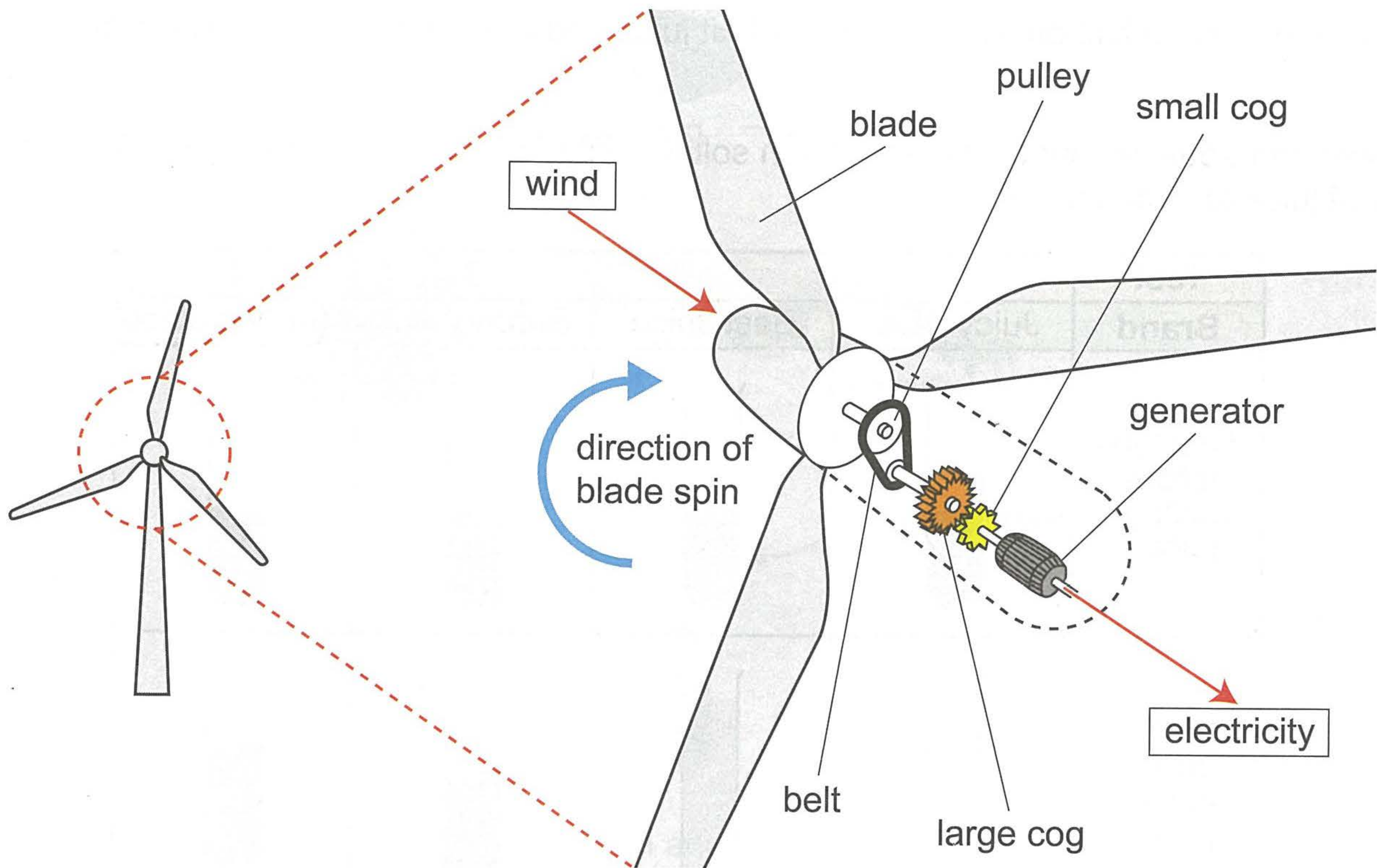
Test	1	2	3	4
Brand	Juicy Plus	Best Juice	Sunny Juice	Morning Juice
Test tube before adding juice				
Test tube after adding juice				

Mark’s test was not a fair test.

What should he have done to make this a fair test?

- (A) used test tubes of the same length
- (B) used a greater amount of juice in each test tube
- (C) used the same amount of juice in each test tube
- (D) used a smaller amount of black solution in each test tube

28. The diagram shows parts of a wind-powered generator which produces electricity.



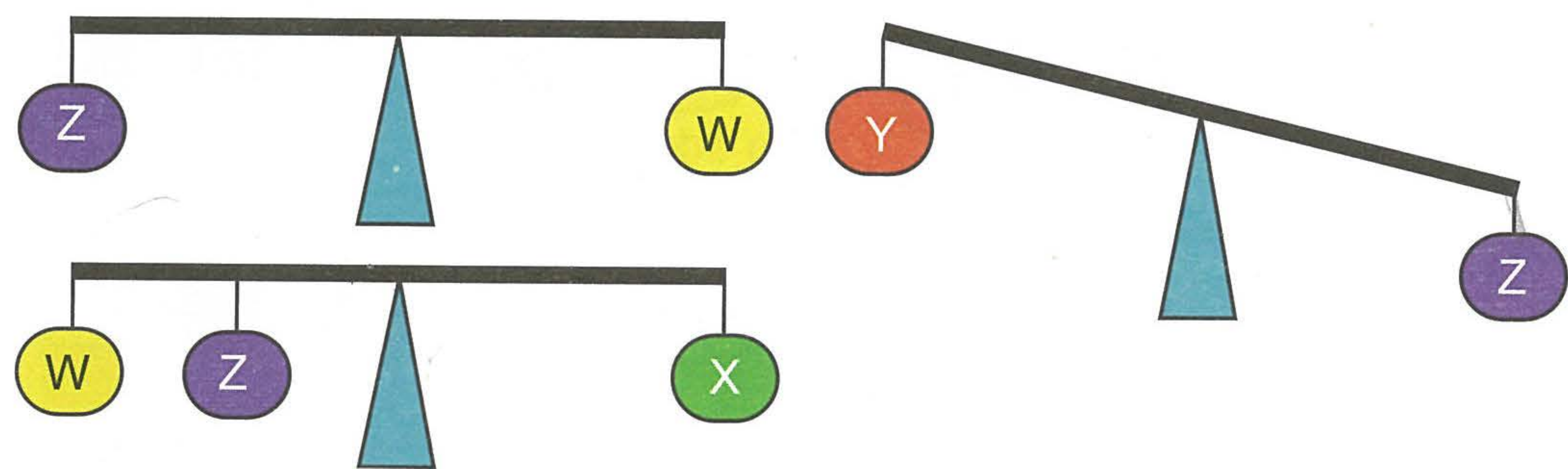
Wind pushes the blades causing them to spin and turn the pulley. The pulley moves the belt which drives the large cog. The large cog turns the small cog, which turns the generator.

Which part of this wind-powered generator would spin the fastest?

- (A) blade (B) pulley (C) large cog (D) small cog

29. A balance beam can be used to compare the masses of two or more objects.

The diagrams show four objects, W, X, Y and Z on some balance beams.



Which object has the greatest mass?

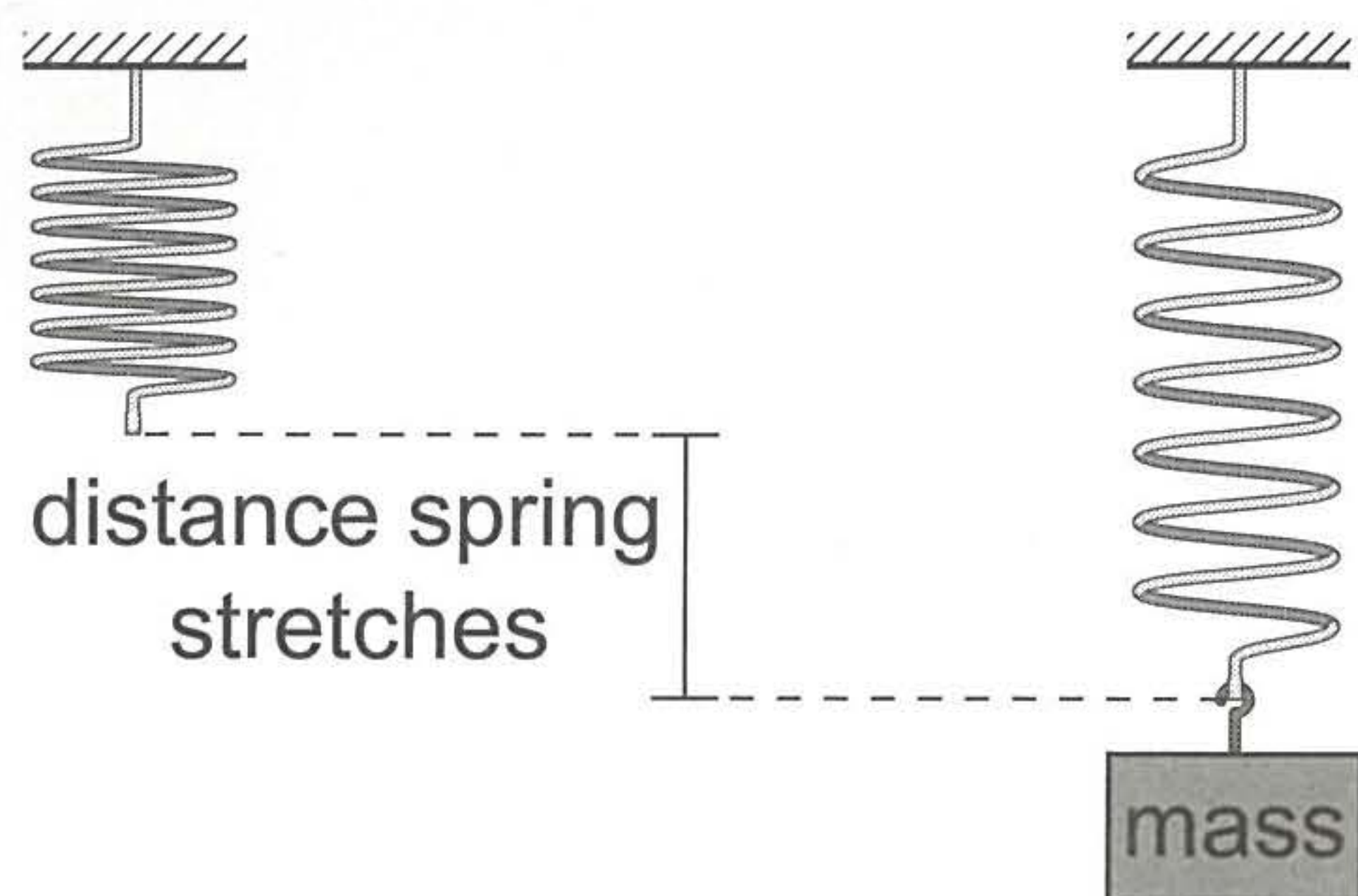
- (A) W (B) X (C) Y (D) Z

30. When a mass is attached to a spring, the spring stretches.

Michelle has a spring that is 200 mm long with no mass attached.

She attaches different masses to this spring and measures the distance that the spring stretches for each mass as shown in the diagram.

The table shows her results.



Attached mass (g)	Distance that the spring stretches (mm)
0	0
100	4
200	8
300	12
400	16

How long is the spring with a mass of 600 g attached?

- (A) 20 mm (B) 24 mm (C) 220 mm (D) 224 mm

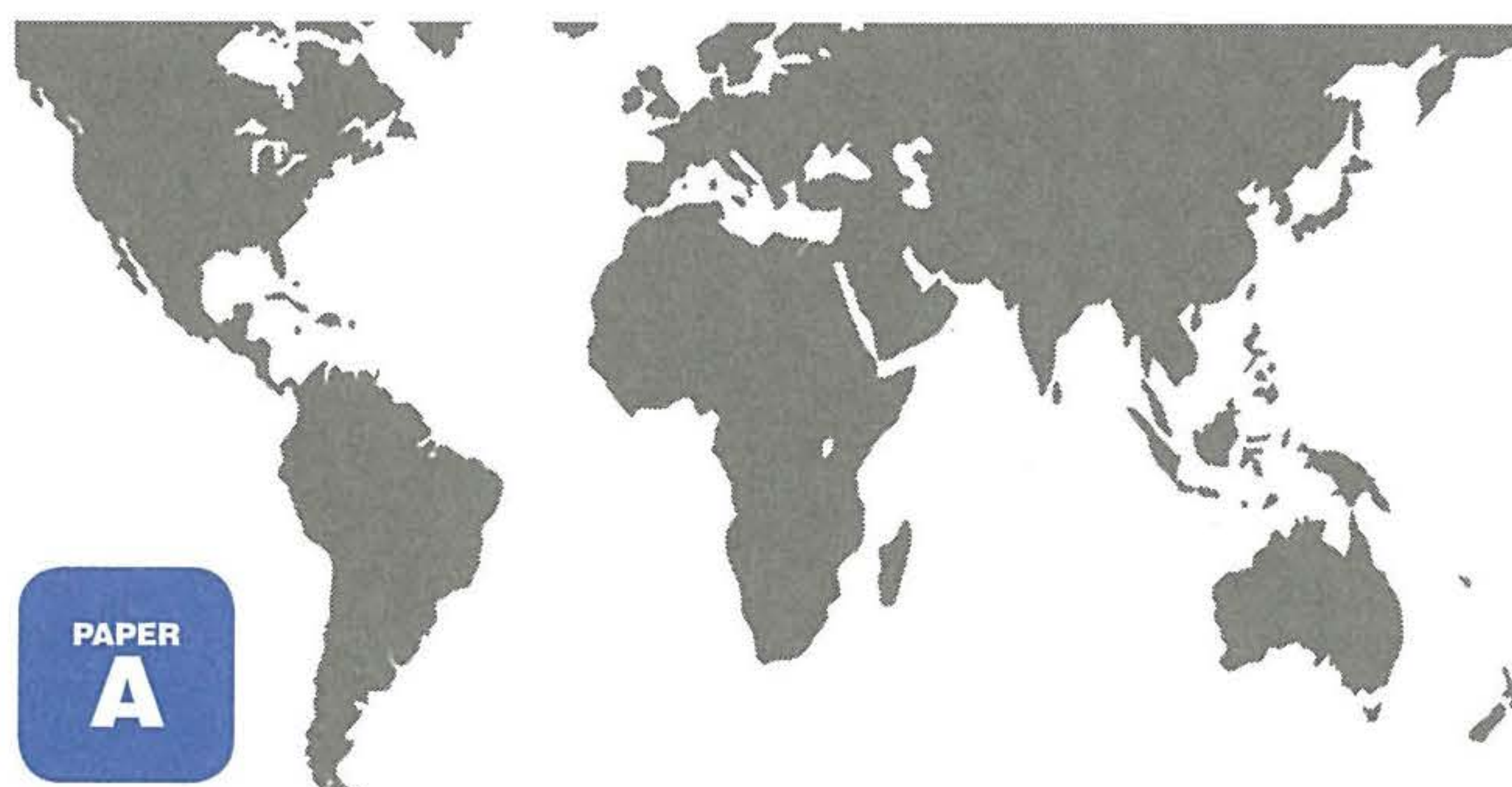
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Australia	Year 3
Brunei	Primary 3
Indonesia	Year 4
Malaysia	Standard 3
New Zealand	Year 4
Pacific	Year 3
Singapore	Primary 2
South Africa	Grade 3



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2011 Science Answer Keys

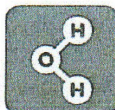
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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
1	A	B	C	B	C	C	A	A	B
2	B	B	D	B	B	D	C	A	A
3	A	A	B	C	C	D	D	C	A
4	C	C	D	D	C	B	C	B	A
5	A	A	A	A	D	C	C	A	D
6	B	C	D	A	C	A	B	C	C
7	C	B	A	D	A	D	C	C	D
8	C	A	D	A	C	C	A	A	C
9	A	B	C	C	B	B	B	C	B
10	D	A	C	D	C	B	B	B	C
11	D	D	D	A	B	B	C	B	D
12	B	B	B	D	D	B	D	C	C
13	C	A	D	B	A	D	A	A	A
14	A	D	C	A	D	C	D	C	C
15	B	A	A	A	B	C	C	B	C
16	D	C	C	C	D	C	C	B	C
17	D	C	C	B	D	D	B	A	C
18	C	D	B	D	C	D	B	A	B
19	A	C	D	D	A	B	C	C	B
20	D	C	D	C	C	C	C	C	C
21	D	B	C	B	C	B	B	D	C
22	C	D	D	C	B	D	C	C	A
23	D	A	A	A	D	A	D	A	C
24	B	A	B	B	A	C	B	B	C
25	C	D	A	D	C	D	D	D	A
26	B	B	C	B	D	D	C	D	B
27	C	A	A	D	A	B	D	B	C
28	D	D	B	B	B	C	D	C	B
29	B	C	D	C	C	D	B	C	C
30	D	D	B	B	C	C	C	D	D

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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
31			B	B	B	C	C	C	B
32			A	A	C	B	B	B	D
33			D	C	C	C	B	C	C
34			A	B	C	A	C	B	B
35			A	C	D	C	A	D	C
36			B	B	A	C	B	B	B
37			C	B	A	A	C	C	A
38			D	D	B	B	B	C	D
39			C	B	D	C	B	D	A
40			B	C	B	C	C	C	D
41					C	C	D	C	C
42					C	A	B	D	A
43					C	C	C	C	B
44					D	C	A	D	B
45					C	C	D	D	A

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